

The Training will begin shortly

Microsoft GitHub Copilot Masterclass: Build Faster, Code Smarter

Harsh Sharma
NetCom Learning

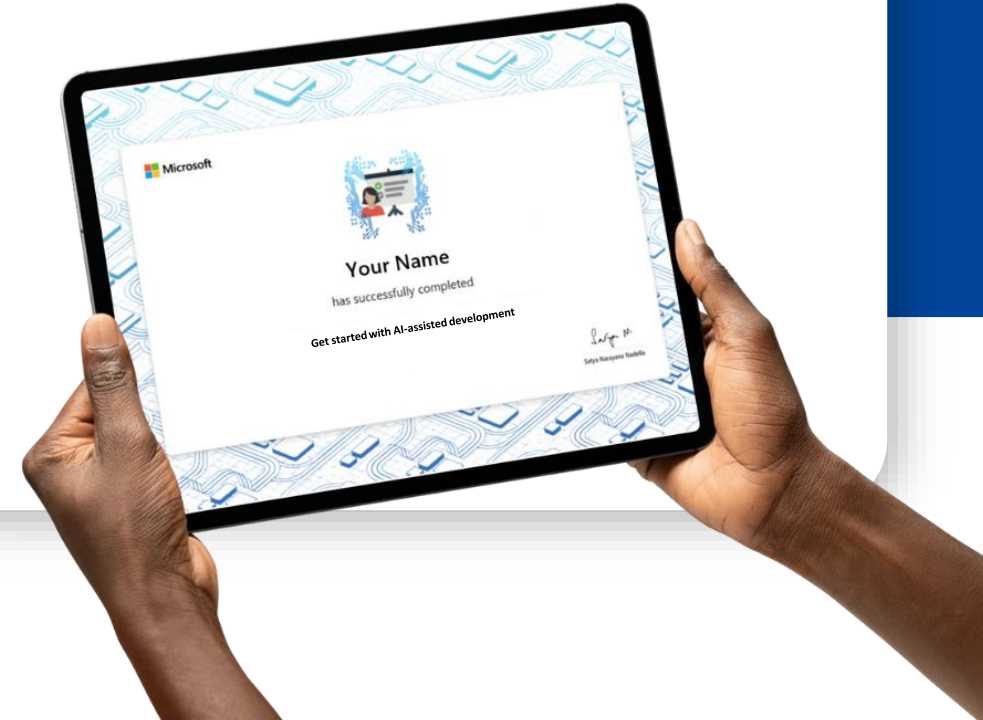
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**Hello,
Thank you for joining me Today!**

Harsh Sharma
NetCom Learning



Agenda:

- ➔ Introduction to AI-Powered Development with GitHub Copilot
- ➔ Understanding How GitHub Copilot Integrates into Modern Development Environments
- ➔ Best Practices for Writing Effective Prompts & Code
- ➔ Enhancing Code Quality, Productivity using GitHub Copilot
- ➔ How to Scale GitHub Copilot Adoption Across Teams: Tools, Metrics & Success Strategies
- ➔ Real-World Use Cases & Enterprise Development Scenarios
- ➔ Q&A Session + Next Steps with the Instructor

Introduction to AI-Powered Development with GitHub Copilot



The AI Development Shift

Artificial intelligence is fundamentally reshaping how developers write, review, and ship code

The Old Way

-  Manual research & Stack Overflow lookup
-  Repetitive boilerplate written from scratch
-  Slow onboarding to unfamiliar codebases
-  Documentation written last (or never)

The AI Way

-  Inline suggestions as you type
-  Scaffolding generated from a comment
-  Explain any codebase instantly
-  Docs, tests, and reviews — all AI-assisted

GitHub Copilot is the most widely adopted AI coding tool — used by over 1.8 million developers worldwide.

What is GitHub Copilot?

An AI pair programmer built by GitHub & OpenAI, embedded directly in your IDE

What Copilot Is

- ✓ AI-powered code completion built into your IDE
- ✓ Powered by OpenAI Codex & GPT-4 models
- ✓ Trained on billions of lines of public code
- ✓ Suggests whole lines or entire functions
- ✓ Understands context from comments & open files

What Copilot Is Not

- ⚠ Not a replacement for developers
- ⚠ Not always 100% accurate — always review
- ⚠ Not responsible for approvals or decisions
- ⚠ Not a substitute for security governance
- ⚠ Not autonomous — human-supervised always

Copilot assists humans — it does not replace human judgment or accountability.

GitHub Copilot Modern Development Environments



Setting Up GitHub Copilot in Your IDE

Get up and running in minutes — across all major development environments

01



Choose a Plan

Sign up at github.com/copilot.
Individual free trial — 30 days.
No credit card required.

02



Install Extension

Search 'GitHub Copilot' in your
IDE marketplace and click
Install.
Reload after installation.

03



Authenticate

Sign in with your GitHub
account.
Accept the license agreement.
MFA supported for Enterprise.

04



Start Coding

Write a comment describing
what
you need — Copilot suggests
code.
Accept suggestions with Tab.

Supported Languages & Development Environments

Programming Languages

- | | |
|------------------|-----------------------------|
| ✓ Python | ✓ JavaScript / TypeScript |
| ✓ Java | ✓ C / C++ / C# |
| ✓ Go / Rust | ✓ Ruby / PHP |
| ✓ Swift / Kotlin | ✓ Shell / Bash / PowerShell |

IDEs & Editors

- </> Visual Studio Code
- </> Visual Studio 2022
- </> JetBrains Suite (IntelliJ, PyCharm...)
- </> Neovim
- </> Xcode (via extension)
- </> Azure Data Studio
- </> GitHub.com web editor
- </> GitHub CLI

Where Copilot Fits in Your Development Workflow

Copilot is embedded at every stage — not just code completion



Plan

Write a comment or issue description — Copilot helps draft implementation steps and pseudocode.



Code

Inline suggestions for functions, classes, and entire files as you type, in any language.



Test

Auto-generate unit tests, integration tests, and edge-case scenarios from your function signature.



Review

Explain code, generate PR summaries, and flag potential bugs before merging.



Document

Write JSDoc, docstrings, and README sections from your existing code in seconds.



Iterate

Refactor, simplify, and improve code quality with /fix, /simplify, and inline suggestions.

Best Practices for Writing Effective Prompts & Code



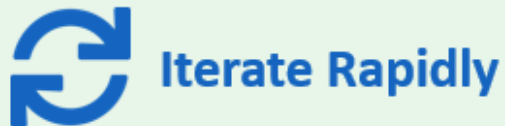
The 4 Principles of Effective Prompt Engineering



Include language, framework, expected inputs/outputs, and edge cases in your comment prompts.



Open related files and keep imports visible — Copilot reads your open tabs for richer context.



Accept partial suggestions, adjust the prompt, and re-prompt. Use Ctrl+Enter for multiple options.



Treat every suggestion like a peer's code — read, test, and validate before committing to production.

Prompt Examples — Vague vs. Specific

✗ Vague Prompt

```
// sort a list
```

```
// make API call
```

```
// validate input
```

```
// parse JSON
```

✓ Specific Prompt

```
// Sort list of dicts by 'age' key  
ascending, Python, return new list
```

```
// GET /api/users/{id} using axios in JS,  
handle 404 & 500 errors
```

```
// Validate email with regex in JS, return  
bool, RFC 5322 compliant
```

```
// Parse JSON string safely in TypeScript,  
return typed interface or null
```

Specific prompts = better suggestions. Always include language, libraries, and I/O expectations.

Copilot Chat — Slash Commands Reference

Use these commands in the Copilot Chat panel to go beyond code completion

Command	What it does	Example use
/explain	Explains selected code in plain English	Why is this function using recursion?
/fix	Suggests a fix for the bug in selection	The loop breaks early — can you fix it?
/tests	Generates unit tests for selected code	Write Jest tests for this auth function
/doc	Writes JSDoc / docstring comments	Add documentation to this class
/simplify	Refactors complex code for readability	This function is hard to read — simplify it
/new	Creates a new file, class, or module	Create a React modal component
/help	Lists all available Copilot Chat commands	What can Copilot Chat do?

Enhancing Code Quality & Productivity using GitHub Copilot



Developer Workflow: Before vs. After Copilot

How AI assistance transforms your daily development experience

Before Copilot	After Copilot
Search Stack Overflow for syntax	Copilot suggests inline — no tab switch
Write boilerplate code from scratch	Scaffolding generated in one keystroke
Manually write every test case	Full test suite drafted in seconds
Context-switch to documentation	Ask Copilot Chat for instant answers
Slow onboarding to new codebases	Explain any code block with one click
PR descriptions written manually	Auto-generated PR summaries from your diff

Teams using GitHub Copilot complete tasks up to 55% faster and report significantly higher focus.

How Copilot Actively Improves Code Quality

AI assistance goes beyond speed — it helps you write more reliable, maintainable code



Test Coverage

Automatically generates unit tests, edge cases, and mocks — reducing gaps in your test suite.



Bug Detection

Copilot Chat's `/fix` command identifies root causes and suggests targeted, context-aware fixes.



Code Refactoring

Use `/simplify` to reduce complexity and improve readability without changing core behavior.



Code Review Aid

Ask Copilot to explain unfamiliar code and flag potential issues before peer review.



Documentation

`/doc` generates accurate JSDoc and docstrings from existing code — no more undocumented functions.



Security Awareness

Copilot flags common vulnerabilities (SQL injection, XSS) and suggests safer coding patterns.

Scaling GitHub Copilot Across Teams



GitHub Copilot Plans — Choosing the Right Tier

Select the plan that fits your individual, team, or enterprise scale

Individual

\$10/mo

per user / month

- ✓ Personal IDE usage
- ✓ VS Code, JetBrains, Neovim
- ✓ Copilot Chat in IDE
- ✓ 30-day free trial

Business

\$19/mo

per user / month

- ✓ All Individual features
- ✓ Org-level policy controls
- ✓ Audit logs & IP indemnity
- ✓ SSO & team management

Enterprise

\$39/mo

per user / month

- ✓ All Business features
- ✓ Copilot in GitHub.com
- ✓ Fine-tuning on private code
- ✓ PR summaries & reviews

Enterprise plan includes fine-tuning on private code and IP indemnity — ideal for regulated industries.

Enterprise Adoption Roadmap

A structured 5-stage approach to rolling out Copilot across your entire engineering organization

- 1 Pilot:** Select 10–20 enthusiastic power users across different teams. → *Collect feedback, identify blockers, and build internal champions.*
- 2 Rollout:** Expand to full teams; run enablement workshops and training sessions. → *Establish consistent usage patterns and share early wins broadly.*
- 3 Governance:** Define acceptable use policies, data handling rules, and review standards. → *Ensure compliance, IP safety, and responsible AI practices org-wide.*
- 4 Optimize:** Track suggestion acceptance rate, PR cycle time, and developer NPS. → *Identify teams with low adoption and provide targeted coaching.*
- 5 Scale:** Full org-wide deployment with a Center of Excellence (CoE) for support. → *Continuous improvement culture; Copilot embedded in all workflows.*

Measuring Success — Key Metrics & Strategies

What to track, celebrate, and iterate on as you scale Copilot adoption



Suggestion Acceptance Rate

% of Copilot suggestions accepted by developers. Target: >30% signals strong daily utility and trust.



PR Cycle Time

Time from first commit to merge. Copilot-enabled teams typically see 20–35% reduction.



Test Coverage Delta

Increase in automated test coverage. Track before/after to quantify quality improvements.



Developer NPS

Measure developer satisfaction. High NPS = Copilot is genuinely improving daily experience.

Change Management — Driving Team Adoption

Getting your engineers, managers, and leadership aligned behind GitHub Copilot



Identify Champions

Recruit enthusiastic early adopters in each team to lead by example, answer questions, and support peer onboarding.



Celebrate Early Wins

Share concrete stories of productivity gains, bugs caught early, and time saved across the whole organization.



Structured Onboarding

Use NetCom's GitHub Copilot training curriculum for consistent, role-based enablement at scale — developers, leads, and architects.



Measure & Iterate

Review acceptance rate and PR cycle time monthly. Use data to coach low-adoption teams and refine your policies.

Real-World Use Cases & Enterprise Development Scenarios



GitHub Copilot in Action — 6 Key Use Cases



Boilerplate & Scaffolding

CRUD endpoints, class scaffolding, config files — generated from a single descriptive comment.



Unit & Integration Tests

Auto-generate Jest, pytest, or JUnit tests including edge cases, mocks, and setup boilerplate.



Debugging & Root Cause

Describe the bug in plain English; Copilot pinpoints the root cause and suggests a targeted fix.



Data Pipelines & SQL

Pandas, SQL queries, ETL logic, and data transformation scripts written from natural language.



Infrastructure as Code

Terraform, Bicep, and CloudFormation templates generated from plain-text architecture descriptions.



Legacy Code Modernization

Explain, refactor, and add tests to existing legacy codebases — accelerating modernization projects.

Measurable Business Impact

Quantified productivity, quality, and satisfaction gains from GitHub Copilot adoption

55%

Faster task completion

74%

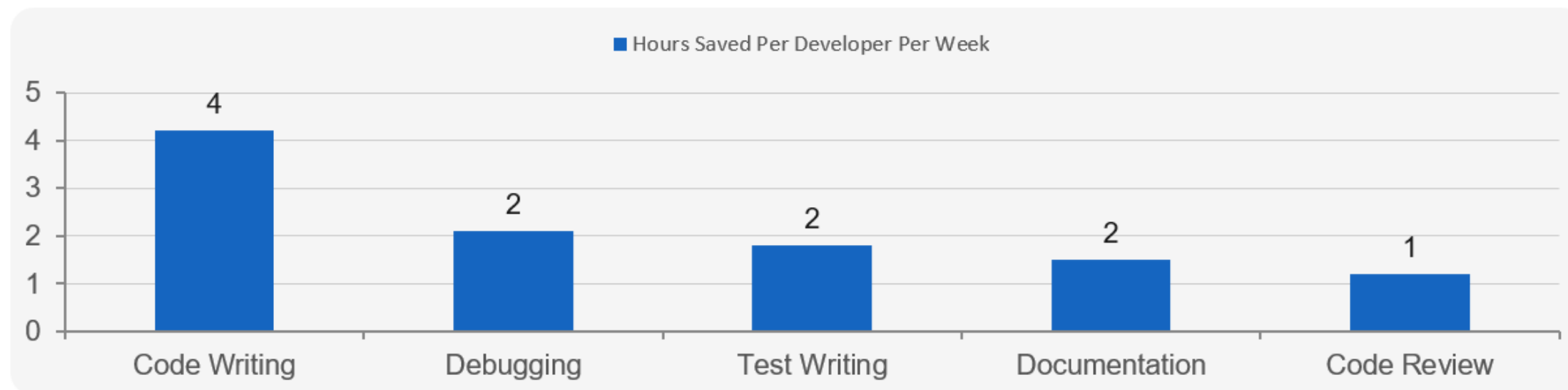
Developers feel more focused

88%

Report higher productivity

3x

More code per session



Demo





The Next Step:

**Continue your Learning &
Get Microsoft Certified**



Getting Microsoft Certified:

What are the benefits of certification?

Stand out

Certified employees **get greater recognition** of skills due to validation¹

Earn more

23% of Microsoft certified technologists **earn up to 20% more**²

Advance

49% believe cloud certifications **increase employability**³

¹ [2017 IDC-Microsoft, Cloud Skills and Organizational Influence: How Cloud Skills Are Accelerating the Careers of IT Professionals](#) white paper

² [2017 Pearson VUE, Value of Certification](#) white paper

³ 2017 [Nigel Frank International Microsoft Technology for Business Salary Survey](#)

Broadcast certified skills

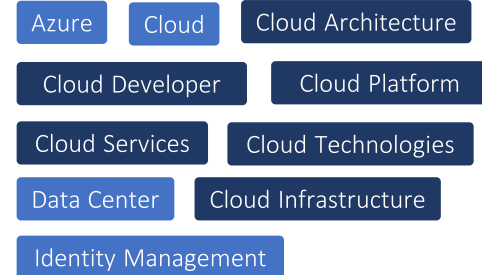


Instant employer verification



This badge is issued by Microsoft

Skills



Unlock the value of **Microsoft Open Badges**

- **Social** - help you share your skills on the social platform of your choosing and your own personal communication
- **Discoverable** – helps people find you and employers who may be interested
- **Verified** – official verification from Microsoft of your accomplishment

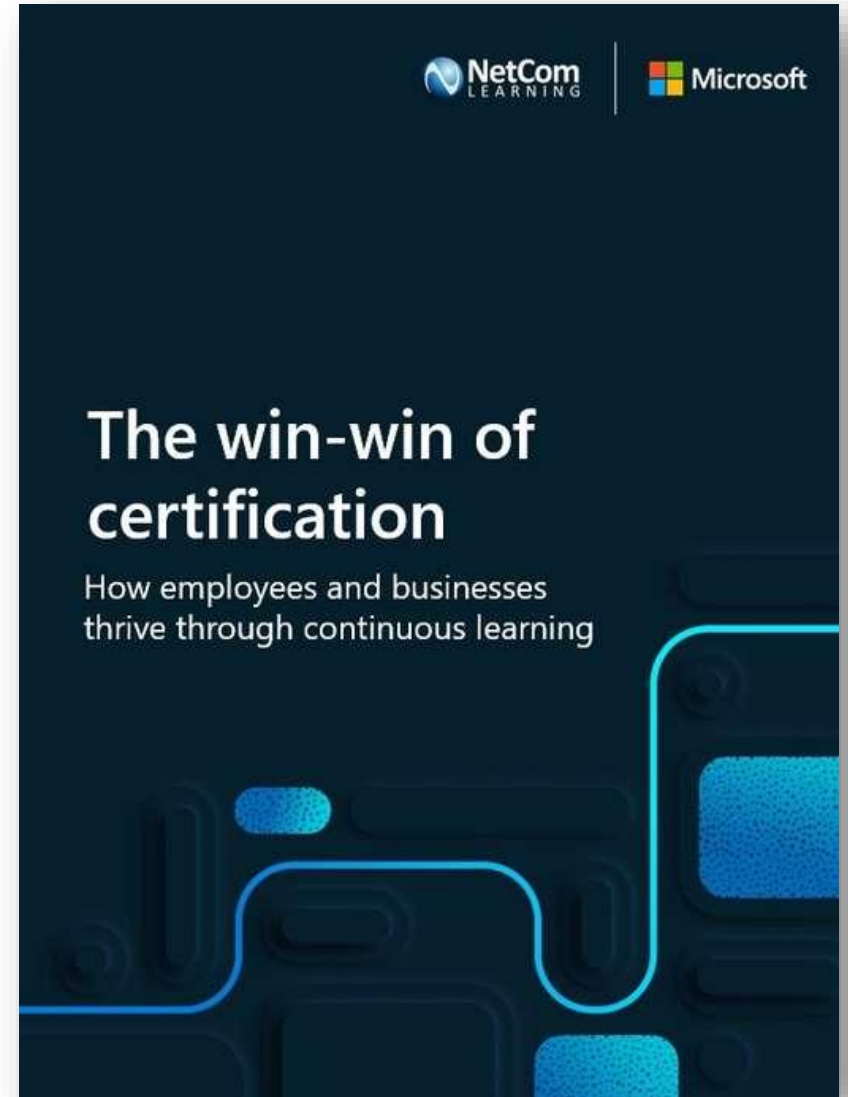
Explore the eBook:

Let the data speak

When organizations maintain a culture of certification and continuous learning, the results are both substantial and measurable: **well-trained and certified teams are almost 20% more productive than other staff.**

Quantify the value of a growth mindset and training. In this eBook, you'll see stats and facts on how certification benefits both employees and organizations, including:

- How training improves timeliness, insight, and business impact
- The personal and career benefits of certification
- The direct relationship between continuous learning and higher pay
- How certification improves productivity, efficiency, and job satisfaction



Download Link is in your Event Handout

Grow your business & career with Official Microsoft **Role-Based** Instructor-led Classroom Training



Key Highlights:

- Highly interactive discussions on your specific **real-world use cases** and business contexts with **Microsoft Certified Trainers** who come with technical subject matter expertise
- Get **180-day** access to extensive virtual labs for practicing and mastering your Microsoft Fundamental Skills.
- Get **90-day** access to the recordings of your Instructor-led classes
- Full Access to **NetCom365 learner portal**
- Opportunities of **FREE repeats**
- Experience **24x7 priority support**

Some facts about Microsoft Certification Renewals

Benefits of renewing your certification

- Maintain your Microsoft Certification with a continuous learning mindset and keep your skills current at no cost.
- Validate your skills with the latest technology updates that affect the job role.
- Renew your certification without having to pay for and pass any exam(s).

Certification renewal requirements

- If you hold a valid Microsoft role-based or specialty certification that expires within approximately six months, you may renew it.
- You will receive an email from Microsoft Certification team 6 months prior to your certification expiration date, once your certification is eligible for renewal.
- To renew your certification, pass the associated assessment on Microsoft Learn *before* your certification expires.
- Passing the renewal assessment on Microsoft Learn is the only way to renew a certification. You cannot renew your certification by passing an exam, even if it's a new or updated version of the exam you initially passed to earn the certification.

Get in touch with NetCom Learning for end-to-end guidance and support on your Microsoft training and certification requirements

Let's Get Trained & Certified

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Questions

Thank You For Attending!

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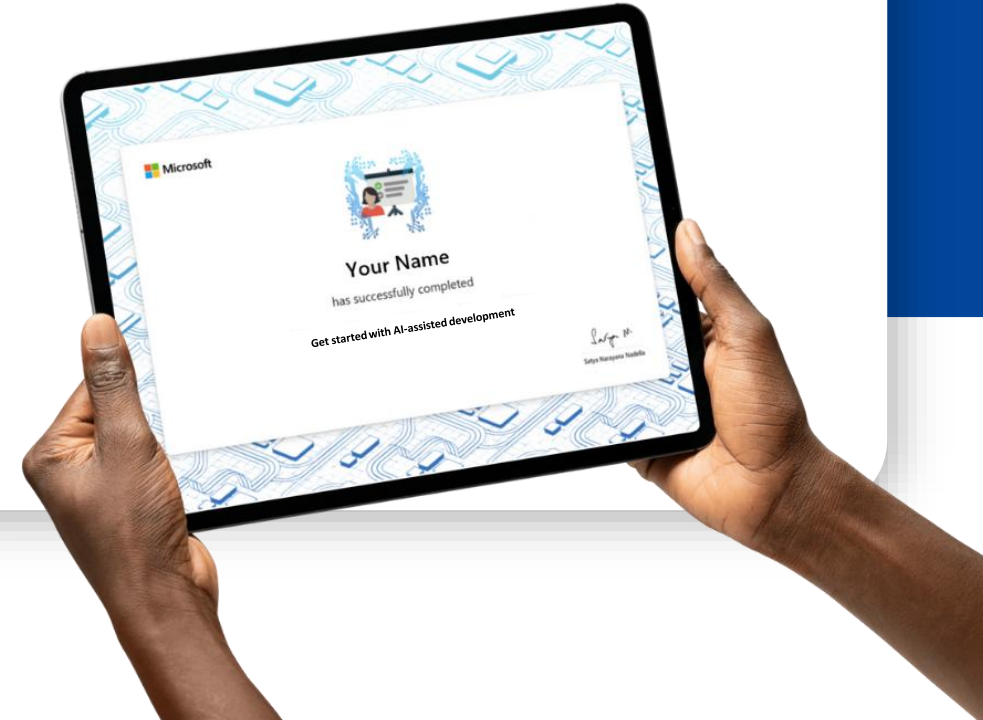
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Thank you

